

SafeVision AI

Real-Time Indoor Fall Detection on the Edge

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Falls Are a Silent Emergency

Falls represent one of the most underreported yet devastating health crises affecting older adults and industrial workers alike. The scale of the problem demands a smarter response.

1 in 4

Adults Over 65

Fall every year, according to CDC data

#1

Injury Death Cause

Among the elderly population nationwide

In warehouses and industrial facilities, slip-and-fall accidents are the **leading category of workplace injury**, costing billions in liability and lost productivity annually.

Why Current Solutions Fail

Solution	Critical Failure
CCTV + human monitoring	No automation, human fatigue
Cloud AI systems	Privacy risk, high latency, expensive
Wearable sensors	Must be worn, forgotten, battery dies
Pressure mats	Limited coverage, no real-time alert

Edge-Based, Privacy-First Fall Detection

SafeVision AI is a fully self-contained fall detection platform engineered to run entirely on affordable edge hardware — no cloud subscription, no privacy compromise, no wearable required.



100% On-Device Processing

All inference and data storage happen locally. No video, keypoint data, or personal information ever leaves the room — privacy preserved by architecture.



Detection in Under 3 Seconds

YOLOv8n-Pose runs at 256×256 inference resolution, achieving real-time throughput with a 5-frame majority vote confirmation window.



Instant Multi-Channel Alerts

Confirmed falls trigger simultaneous email and SMS notifications with an attached snapshot photo, all within approximately 1.5 seconds of detection.

How SafeVision AI Works

SafeVision AI is composed of five tightly integrated Python modules, each responsible for a distinct layer of the pipeline — from raw frame capture through alert delivery and event persistence.



The modular design ensures each component can be tested and upgraded independently. The detection pipeline is non-blocking — alert failures never interrupt frame processing or pose inference.

Tech Stack

Model: YOLOv8n-Pose · **Framework:** PyTorch + OpenCV · **Server:** Flask + Socket.io · **Hardware:** Jetson Orin Nano (JetPack 5.1.4) · **Alerts:** SendGrid + Twilio · **DB:** SQLite

Detecting Falls Through Skeleton Analysis

Keypoint Extraction

YOLOv8n-Pose detects **17 COCO body keypoints** per person in each frame. SafeVision focuses on four anatomically significant points to construct the torso midline vector.

1

Anchor Points

Left shoulder, right shoulder, left hip, right hip — forming the torso bounding rectangle.

2

Torso Vector

Midpoint of hips → midpoint of shoulders. Angle measured against vertical using `arctan2`.

Fall Confirmation Logic

```
torso_vec = shoulder_mid - hip_mid  
angle = arctan2(|torso_vec_x|,  
               |torso_vec_y|)
```

Upright person → $0^\circ - 30^\circ$

Bending/sitting → $30^\circ - 44^\circ$

FALL confirmed → $\geq 45^\circ$
in 3 of 5 frames

❗ A majority vote across 5 consecutive frames eliminates false positives from bending, sitting, or picking objects up — while keeping total detection latency under 3 seconds.

Real-Time Monitoring from Any Browser

SafeVision's Flask + SocketIO web dashboard delivers live situational awareness to any device on the local network — no app installation required.

Live Status Bar

Displays current FPS, torso angle, total falls detected today, and system uptime — updated every frame via WebSocket push.

Skeleton Video Feed

Real-time stream with green keypoints and bones during normal activity. Skeleton turns **red** and a blinking **FALL DETECTED** badge appears on confirmed events.

Event Log

Timestamped entries for every fall, including the exact torso angle at confirmation — e.g., 48.6° and 71.3° in live test sessions.

Angle History Chart

Continuous plot of torso angle over time with the 45° fall threshold line clearly visible, enabling retrospective review of borderline events.

Instant Alerts — Email, SMS, and Local Snapshot

The moment a fall is confirmed, SafeVision AI triggers a coordinated, non-blocking alert sequence designed to notify caregivers within seconds while preserving full pipeline continuity.

1

1 · Snapshot Saved

Frame captured and written to `/snapshots/fall_TIMESTAMP.jpg` on-device storage immediately.

2

2 · Email Sent

SendGrid API delivers snapshot attachment with subject *"SafeVision AI: Fall Detected"* to configured recipients.

3

3 · SMS Sent

Twilio delivers a text notification to the configured phone number within approximately 2 seconds of confirmation.

4

4 · Dashboard + DB

WebSocket pushes a live event log entry; SQLite permanently records timestamp, angle, and snapshot path.



Key design decision: Alerts are fully non-blocking. If email or SMS credentials are not configured, the detection pipeline continues running without any interruption — detection never stops.

Performance Summary

Metric	Target	Achieved
Fall detection speed	≤ 15 frames	3–5 frames
Alert latency	≤ 2 seconds	~1.5 seconds
Inference resolution	640×640	256×256 (optimized)
Detection accuracy	F1 > 0.90	Confirmed on test images
System uptime	$\geq 99\%$	Stable, auto-restarts via systemd
Privacy	100% on-device	✓ Zero data leaves device

Engineering Challenges Overcome

- Generic PyTorch had no GPU support → installed Jetson-specific compiled wheel
- OpenCV Qt display errors → switched to headless mode with browser dashboard
- `arccos` angle calculation was numerically unstable → replaced with `arctan2`
- 15-frame window too slow at low FPS → reduced to 5-frame majority vote

Roadmap — What Comes Next

SafeVision AI is designed as a platform, not a one-time project. Four development phases extend the system from single-person fall detection into a comprehensive indoor safety intelligence layer.



Phase 2 · Hazard Detection

YOLOv8n object detection running alongside pose estimation. Detects wet floors, obstacles, open cabinets, and loose cables. Enables context-aware alerts: *"Fall detected near wet floor sign."*



Phase 3 · Behavioral Fusion

Per-second classification of Safe / Hazard Present / Fall states. Detects stumbling and near-falls before impact. Multi-person tracking with person re-identification for crowded environments.



Phase 4 · Scale and Deployment

Multi-camera support with a unified central dashboard. TensorRT engine export for an additional 2× GPU speedup. Fine-tuning on real indoor fall datasets and optional encrypted cloud backup of event logs only — never video.

SafeVision AI — Key Takeaways

What Was Built

A fully working real-time fall detection system running entirely on a **edge device** — no cloud, no subscription.

Live web dashboard with video streaming, event log, and torso angle history chart accessible from any browser on the local network.

Instant email and SMS alerts with snapshot evidence, delivered in approximately **1.5 seconds** with **zero data leaving the device**.

What Was Learned

Edge AI deployment requires **hardware-specific optimization at every layer** — PyTorch wheels, inference resolution, threading, and frame skipping all matter.

Pose-based fall detection is fundamentally more reliable than pixel-difference or motion-based approaches for distinguishing intentional and accidental posture changes.

Small architecture decisions — `arctan2` vs `arccos`, majority vote vs single-frame — have **outsized impact** on real-world reliability.

"SafeVision AI demonstrates that production-grade safety AI doesn't require the cloud — just the right edge hardware, the right model, and thoughtful software engineering."